Structure-Constrained Bidirectional Cross-Attention for Self-Supervised Molecular Representation Learning

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Introduction

Challenge

- ADMET prediction requires integrating both atomic signals and substructural context.
- Atom- or substructure-level graphs alone are insufficient for comprehensive modeling.
- Atoms within the same substructure can exhibit different contributions to molecular activity.

Approach

- Create a unified representation combining atom- and substructure-level graphs.
- Fuse these dual views via membership-constrained bidirectional cross-attention.
- Learn differential atom contributions within each substructure.

Datasets

Pre-training

- ZINC250k[1]: Large and diverse molecular dataset covering broad chemical space, used to learn generalizable representations.

Fine-tuning

- MoleculeNet[2]: Classification & Regression task(8:1:1 split)
- MoleculeACE[3]: Activity cliffs benchmark (~35k molecules across 30 targets)

Table 1. Details of MoleculeNet Benchmarks Datasets

Dataset	#Tasks	#Samples	Type	Metric
BBBP	1	2,040	Classification	ROC-AUC
ClinTox	2	1,478	Classification	ROC-AUC
Tox21	12	7,831	Classification	ROC-AUC
SIDER	27	1,427	Classification	ROC-AUC
BACE	1	1,513	Classification	ROC-AUC
FreeSolv	1	639	Regression	RMSE
ESOL	1	1,127	Regression	RMSE
Lipo	1	4,200	Regression	RMSE

Methods

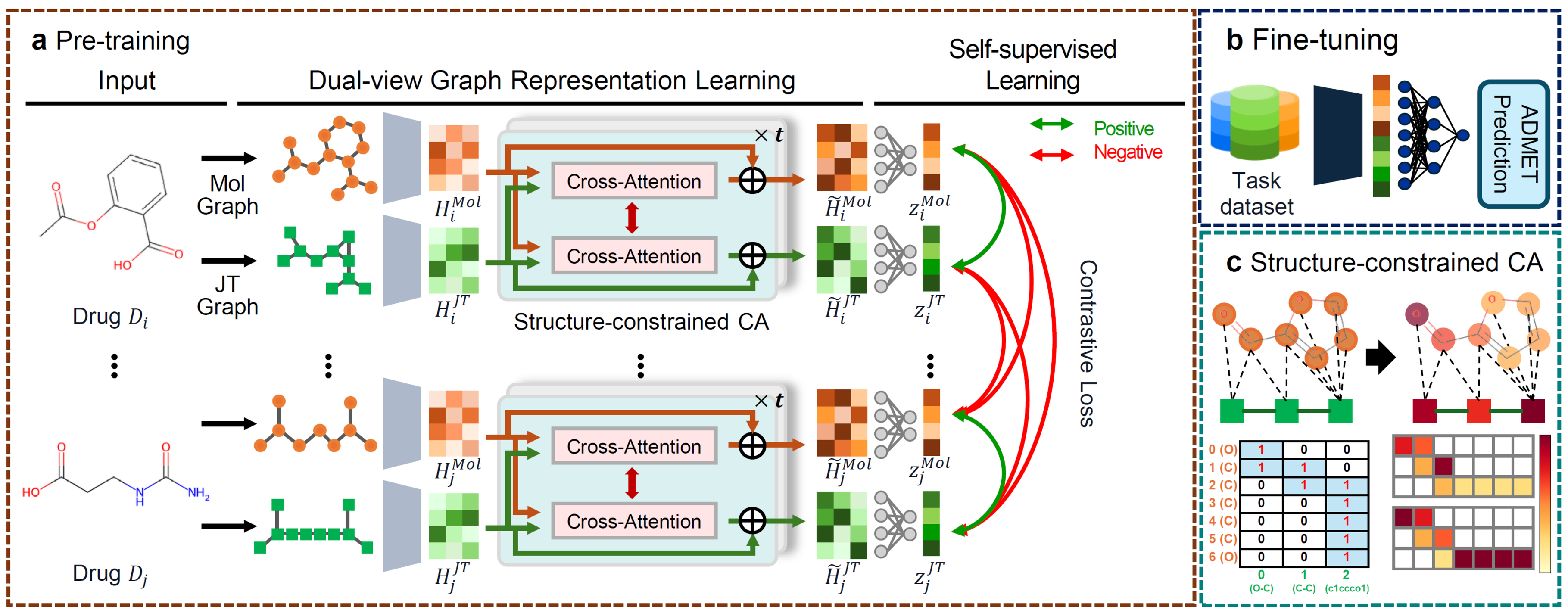

Figure 1. Overview of Our Framework

a. Pre-training

- **Input:** Drug D_i → Mol graph (G_i^{Mol}) & JT graph (G_i^{JT})
- **Dual-view Graph Representation Learning:** GATv2 encodes Mol/JT graphs into atom- and substructure-level embeddings. Structure-constrained CA updates both views using atom–substructure mappings.
- **Self-supervised Leering:** Contrastive loss aligns same-drug Mol/JT embeddings and separates different-drug embeddings.

b. Fine-tuning

- **Task Datasets:** Labeled molecular property datasets
- **ADMET Prediction:** Pooled Mol/JT embeddings → MLP → property prediction

c. Structure-constrained Cross-attention

- **Mapping:** Atom–substructure membership defines valid pairs.
- **Masked Attention:** Only mapped pairs attend; invalid pairs are masked out.

Results

1. Overall Performance on MolecueNet Benchmarks

- The proposed model shows robust performance across MoleculeNet classification and regression tasks.
- Performance gains indicate the benefit of jointly modeling atom-level and substructure-level molecular views.

Table 2. Performance on the MoleculeNet ADMET benchmarks (random split, 10 seeds)

Model	Classification task (ROC-AUC↑)						Regression task (RMSE↓)		
	BBBP	BACE	ClinTox	SIDER	Tox21	ToxCast	ESOL	FreeSolv	Lipo
GCN	0.867(0.039)	0.658(0.070)	0.538(0.118)	0.563(0.020)	0.716(0.017)	0.670(0.015)	1.814(0.120)	3.099(0.317)	0.967(0.031)
GAT	0.865(0.038)	0.645(0.065)	0.534(0.114)	0.570(0.021)	0.733(0.022)	0.657(0.028)	1.836(0.109)	2.748(0.426)	0.960(0.027)
GIN	0.882(0.036)	0.696(0.062)	0.602(0.096)	0.563(0.021)	0.784(0.017)	0.663(0.022)	1.444(0.117)	2.924(0.404)	0.888(0.033)
D-MPNN	0.924(0.021)	0.885(0.022)	0.841(0.075)	0.606(0.036)	0.837(0.013)	0.746(0.016)	0.594(0.072)	1.182(0.192)	0.569(0.032)
AttentiveFP	0.912(0.033)	0.868(0.034)	0.906(0.034)	0.628(0.035)	0.853(0.013)	0.762(0.009)	0.632(0.091)	1.270(0.261)	0.616(0.025)
ChemBERTa-v2	0.934(0.023)	0.884(0.023)	0.765(0.069)	0.647(0.034)	0.840(0.017)	0.744(0.008)	0.771(0.077)	1.439(0.232)	0.617(0.045)
MoLFormer	0.917(0.027)	0.887(0.020)	0.898(0.019)	0.635(0.031)	0.843(0.021)	0.748(0.016)	0.645(0.068)	1.150(0.201)	0.593(0.024)
pretrainGNN (AM)	0.933(0.019)	0.879(0.023)	0.700(0.046)	0.630(0.041)	0.844(0.014)	0.735(0.015)	0.843(0.114)	1.650(0.286)	0.641(0.029)
GPT-GNN	0.917(0.023)	0.866(0.019)	0.670(0.101)	0.604(0.017)	0.833(0.017)	0.718(0.018)	1.054(0.091)	1.942(0.207)	0.765(0.028)
GraphCL	0.926(0.018)	0.860(0.023)	0.721(0.079)	0.620(0.029)	0.839(0.018)	0.746(0.015)	0.901(0.140)	1.528(0.228)	0.658(0.042)
GROVER	0.917(0.017)	0.855(0.022)	0.821(0.066)	0.647(0.025)	0.846(0.011)	0.750(0.007)	0.621(0.082)	1.153(0.202)	0.710(0.028)
MolCLR	0.931(0.027)	0.874(0.023)	0.845(0.063)	0.617(0.034)	0.844(0.016)	0.743(0.010)	0.876(0.103)	1.524(0.501)	0.648(0.026)
iMolCLR	0.921(0.021)	0.841(0.044)	0.849(0.048)	0.616(0.031)	0.840(0.016)	0.734(0.008)	0.878(0.111)	1.741(0.783)	0.661(0.034)
GraphMVP	0.929(0.024)	0.873(0.028)	0.726(0.089)	0.611(0.035)	0.844(0.017)	0.741(0.013)	0.861(0.118)	1.522(0.228)	0.648(0.038)
MPG	0.941(0.016)	0.545(0.051)	0.903(0.043)	0.510(0.020)	0.522(0.029)	0.501(0.013)	0.669(0.087)	0.950(0.170)	0.591(0.034)
Motif	0.933(0.008)	0.881(0.024)	0.722(0.092)	0.615(0.043)	0.843(0.015)	0.743(0.018)	0.889(0.138)	1.652(0.233)	0.644(0.029)
HiMol	0.894(0.013)	0.879(0.024)	0.685(0.145)	0.633(0.030)	0.836(0.021)	0.730(0.017)	0.752(0.085)	1.722(0.751)	0.685(0.024)
MMGX	0.940(0.014)	0.898(0.019)	0.874(0.043)	**0.690(0.037)**	0.770(0.067)	0.783(0.038)	0.623(0.048)	1.343(0.462)	0.586(0.027)
Ours	**0.955(0.013)**	**0.921(0.029)**	**0.921(0.015)**	0.633(0.021)	**0.864(0.016)**	**0.783(0.019)**	**0.543(0.031)**	**0.881(0.194)**	**0.566(0.021)**

***Note**. Values are reported as mean (std). Baselines are grouped as follows: supervised GNNs (GCN, GAT, GIN, D-MPNN, AttentiveFP), SMILES-based pretraining (ChemBERTa-v2, MoLFormer), graph SSL (pretrainGNN, GPT-GNN, GraphCL, GROVER, MolCLR, iMolCLR, GraphMVP), and substructure/motif learning (MPG, Motif, HiMol, MMGX). The best result for each dataset is **bolded**.*

2. Ablation Study

- The full model shows the strongest overall performance, confirming the benefit of dual-view pre-training and structure-constrained bidirectional cross-attention.

Table 3. Ablation study

Variant	Classification task (ROC-AUC ↑)						Regression task (RMSE ↓)		
	BBBP	BACE	ClinTox	SIDER	Tox21	ToxCast	ESOL	FreeSolv	Lipo
Pre-training									
w/o pre-train	0.911(0.022)	0.891(0.011)	0.830(0.036)	0.620(0.031)	0.838(0.016)	0.724(0.009)	0.621(0.024)	1.265(0.401)	0.692(0.011)
Cross-attention depth									
w/o CA (CA=0)	0.920(0.023)	0.869(0.018)	0.842(0.043)	0.617(0.029)	0.834(0.008)	0.725(0.010)	0.608(0.046)	1.515(0.310)	0.605(0.013)
CA=1	0.925(0.018)	0.889(0.015)	0.785(0.071)	0.608(0.028)	0.842(0.009)	0.743(0.008)	0.643(0.021)	1.626(0.655)	0.614(0.020)
CA=2	0.919(0.018)	0.892(0.009)	0.812(0.014)	0.619(0.022)	0.837(0.011)	0.736(0.010)	0.654(0.045)	1.142(0.106)	0.626(0.016)
CA=4	0.918(0.025)	0.883(0.018)	0.822(0.020)	0.601(0.024)	0.845(0.013)	0.736(0.006)	0.636(0.051)	1.398(0.327)	0.607(0.008)
CA=5	0.929(0.004)	0.890(0.022)	0.819(0.077)	0.628(0.017)	0.839(0.018)	0.735(0.010)	0.587(0.023)	1.292(0.285)	0.619(0.017)
Structure constraint									
w/o mask	0.924(0.017)	0.875(0.024)	0.843(0.013)	0.623(0.016)	0.837(0.014)	0.737(0.005)	0.644(0.028)	1.220(0.219)	0.624(0.003)
uniform attn	0.926(0.018)	0.881(0.024)	0.853(0.023)	0.629(0.029)	0.839(0.012)	0.735(0.005)	0.613(0.040)	1.225(0.251)	0.624(0.018)
Attention direction									
JT→Mol only	0.921(0.011)	0.891(0.022)	**0.912(0.017)**	0.597(0.019)	0.844(0.013)	0.740(0.008)	0.598(0.087)	1.212(0.335)	0.610(0.007)
Mol→JT only	0.909(0.028)	0.895(0.028)	0.850(0.012)	0.613(0.026)	0.834(0.010)	0.733(0.007)	0.628(0.008)	1.202(0.151)	0.618(0.024)
Single-view									
JT only	0.922(0.011)	0.883(0.013)	0.863(0.033)	0.607(0.020)	0.839(0.014)	0.732(0.013)	0.677(0.046)	1.266(0.186)	0.606(0.018)
Mol only	0.934(0.014)	0.870(0.021)	0.876(0.041)	0.606(0.017)	0.844(0.008)	0.740(0.007)	0.585(0.020)	1.135(0.163)	0.613(0.004)
Ours (CA=3)	**0.955(0.013)**	**0.921(0.029)**	0.902(0.013)	**0.633(0.021)**	**0.864(0.016)**	**0.783(0.019)**	**0.543(0.031)**	**0.881(0.194)**	**0.566(0.021)**

3. Overall Performance on MoleculeACE Benchmarks

- The proposed model achieves the lowest overall and cliff-specific RMSE, showing robust activity-cliff prediction.

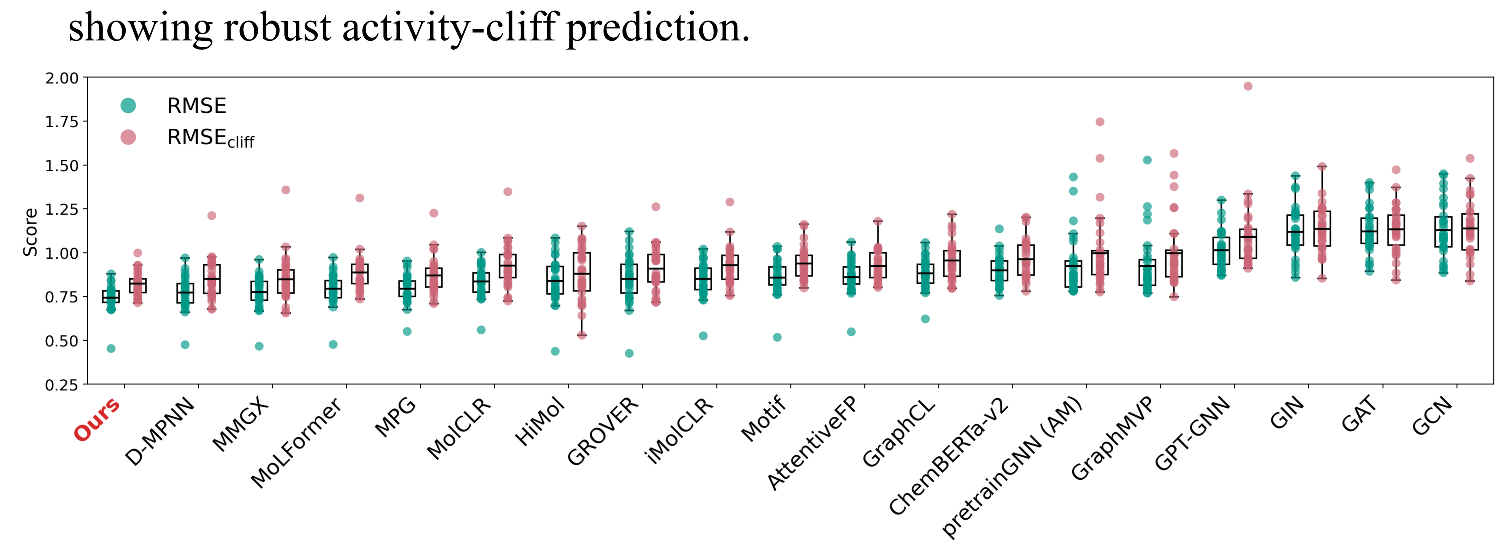

Figure 2. MoleculeACE Benchmark Performance Across 30 ChEMBL Targets

4. Drug Representation Visualization

- Fine-tuned drug embeddings show task-discriminative patterns and cluster by JT.

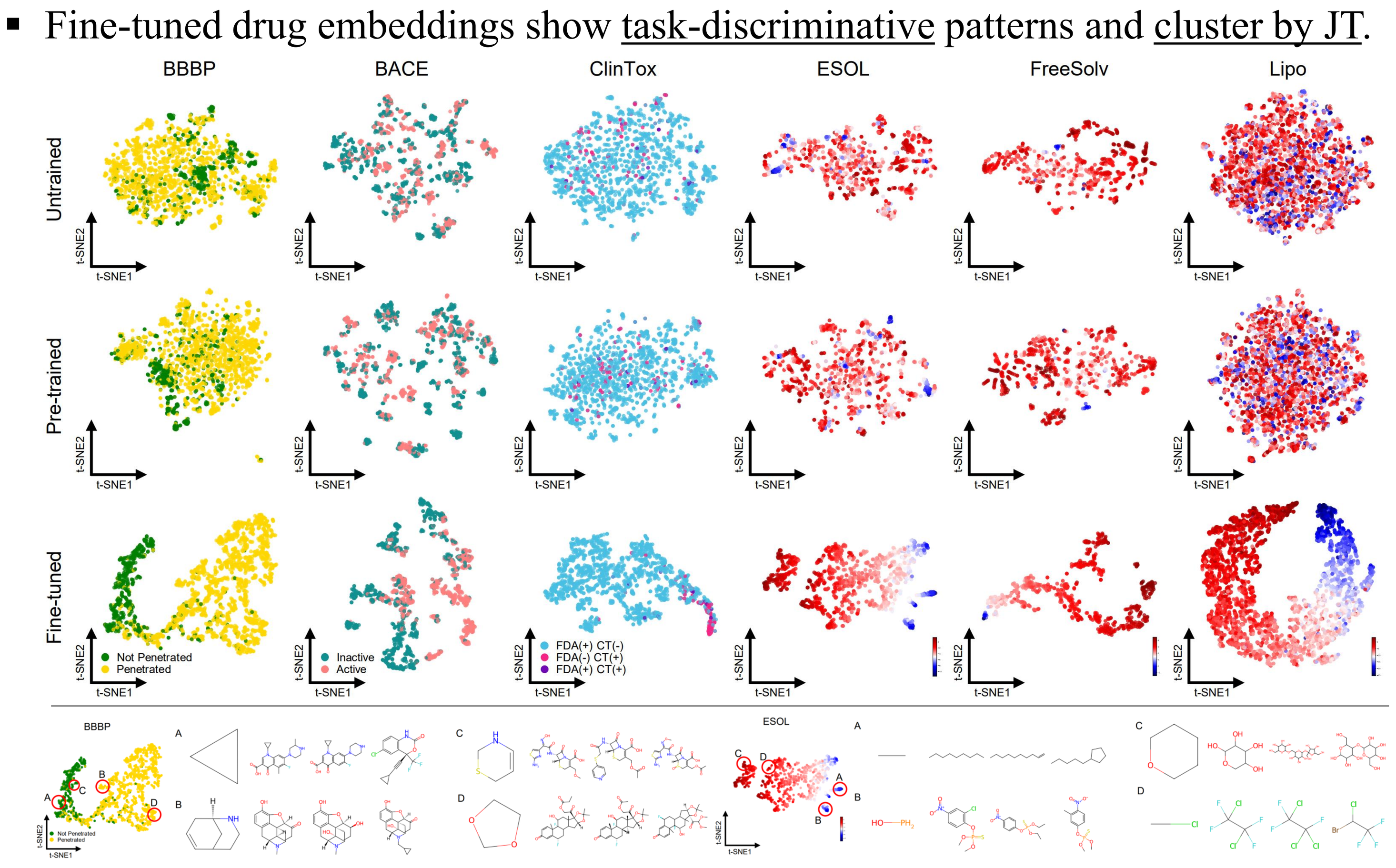

Figure 3. t-SNE Visualization of Drug Representations Across Tasks

5. Cross-attention learns task-specific atom–substructure correspondences

- Cross-attention assigns task-specific weights to the same atom–substructure pairs, enabling intrinsic interpretability without post-hoc attribution.

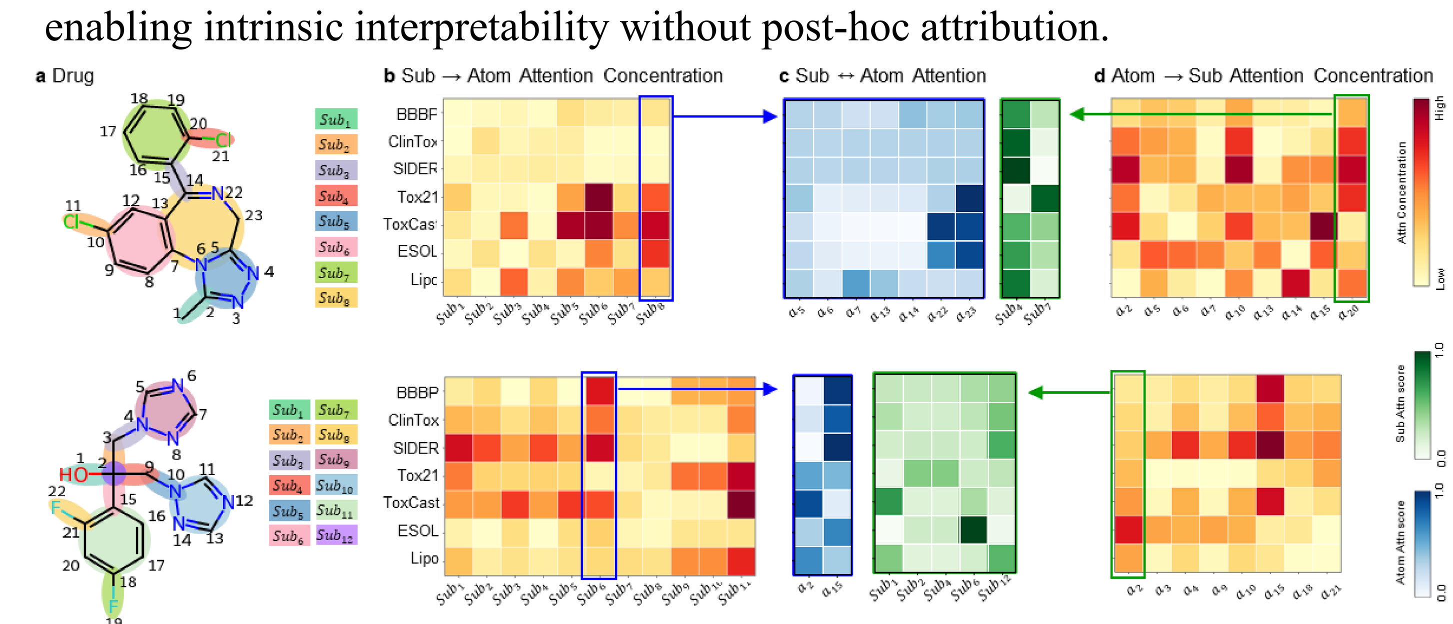

Figure 4. Task-specific CA patterns at the final CA block (CA3) for three representative drugs.

References

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